3.7 Transformations of Graphs of Linear Functions

Name _____ Period ____ Date _____

Algebra I 3.7 · Student sheet

I can…

- describe translations and reflections of linear functions.
- describe stretches and shrinks of linear functions.
- write a function g in terms of f for a transformation.

Vocabulary · fill in the missing word

Parent linear function — f(x) = _____

Translation — f(x) + k _____ up or down; f(x − h) shifts left or right

Reflection — −f(x) in the _____; f(−x) in the y-axis

Stretch / shrink — a · f(x) vertically; f(ax) _____

Part 1 · Desmos transformations

Graph f(x) = x in Desmos. Then graph each function and describe the change.

Function	Change from f(x) = x
f(x) + 2	
f(x − 2)	
−f(x)	
3f(x)	
f(½x)	

Part 2 · Notes: fill in the blanks

f(x) + k moves the graph _____; f(x − h) moves it _____.

−f(x) reflects in the _____; f(−x) reflects in the _____.

a · f(x) with a > 1 is a vertical _____.

Part 3 · Examples (work them with the class)

Example 1

Let f(x) = x − 1 and g(x) = f(x) + 3. Write g(x) and describe the transformation.

Example 2

Let f(x) = 2x and g(x) = f(−x). Write g(x) and describe the transformation.

Example 3

Let $f(x) = x + 4$ and $h(x) = f(3x)$. Write $h(x)$ and describe the transformation.

Example 4

Let $f(x) = x - 3$. Write g, a vertical stretch by a factor of 2 followed by a translation 1 unit down.

Part 4 · Partner practice

Partner 1

Let $f(x) = x$ and $g(x) = x - 5$. Describe the transformation.

Partner 2

Let $f(x) = 3x + 1$. Write $g(x) = -f(x)$.